Seat No: ______________ **Enrollment No:** ____________________

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

B.Tech. Winter 2018 - 19 Examination

Semester: 5 **Date: 27/11/2018**

Subject Code: 03105302 **Time: 10:30 am to 1:00 pm**

Subject Name: Design and Analysis of Algorithm **Total Marks: 60**

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions : **(15)**

1. Which of the given options provides the increasing order of asymptotic complexity of functions f1, f2, f3 and f4?
   f1(n) = 2^n
   f2(n) = n^(3/2)
   f3(n) = nLogn
   f4(n) = n^(Logn)

A) f3, f2, f4, f1
B) f3, f2, f1, f4
C) f2, f3, f1, f4
D) f2, f3, f4, f1

2. Randomized quicksort is an extension of quicksort where the pivot is chosen randomly. What is the worst case complexity of sorting n numbers using randomized quicksort?

A) O(n)
B) O(n log n)
C) $O(n^2)$
D) O(n!)

3. To determine the efficiency of an algorithm the time factor is measured by:

A) Counting micro seconds
B) Counting numbers of key operations
C) Counting numbers of statements
D) Counting Kilobytes of algorithm

4. An all-pairs shortest-paths problem is efficiently solved using:

A) Dijkstra algorithm
B) Bellman-Ford algorithm
C) Kruskal algorithm
D) Floyd- Warshall algorithm

5. An unordered list contains n distinct elements. The number of comparisons to find an element in this list that is neither maximum nor minimum is

A) ϴ (n logn)
B) ϴ (n)
C) ϴ (logn)
D) ϴ (1)

6. Define: Spurious hit
7. State whether this statement is True or False:
   Let $f(n) = 2^{2n}$ and $g(n)= 2^n$ be asymptotically positive functions. Is f(n) = O(g(n))?
8. Reorder the given asymptotic complexity of functions in increasing order : n^{logn} , $n^{2/3}$, $2^{n,}$, $nlog_2n$
9. Write Recurrence equation for Merge Sort.
10. Define : Branch and Bound approach
11. Define : Backtracking
12. Which data structure used in DFS?

13. What is articulation point in graph?
14. Define: MST
15. Define: Asymptotic Notation

Q.2 Answer the following questions. (Attempt any three) **(15)**

A) Explain the Divide and conquer approach in merge sort.

B) Explain how DFS can be used to detect a cycle in a graph? What are the other applications of DFS?

C) With respect to Dynamic Programming, explain in brief the following:
i) Optimal Substructure.
ii) Overlapping Subproblems.

D) Solve the recurrence equation $T(n) = 2T(n/3) + n$ using recursion tree method.

Q.3 A) What is Finite Automata? Explain use of finite automata for string matching with suitable Example. **(07)**

B) Write a algorithm for counting sort and apply on given data: A = { 2,0,3,1,2 } **(08)**

OR

B) Discuss 0-1 knapsack problem. Consider an instance of a knapsack problem with four items and weight carrying capacity as W=5. Weight and benefit of items are given below. Solve this problem using 0-1 Knapsack. **(08)**

Item No.	Weight	Benefit
1	2	3
2	3	4
3	4	5
4	5	6

Q.4 A) Discuss and derive recurrence relation for longest common subsequence problem using Dynamic Programming. Find longest common subsequence of following two strings X and Y using Dynamic Programming. X= cabcba , Y= abcbcba **(07)**

OR

A) Solve Making Change problem using Dynamic Programming. (Denominations: d1=1, d2=4, d3=6). Give your answer for making change of Rs. 8. **(07)**

B) Find Minimum Cost Spanning Tree using Prim's algorithm of the following graph. What is the Time complexity of Prim's algorithm? **(08)**